

UNITED STATES PATENT APPLICATION

APPLICANT / ASSIGNEE: B-Lawz Music LLC

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TITLE OF INVENTION

TECHNICAL FIELD

The present invention relates to adaptive selection systems for artificial intelligence content generation and model training, and more particularly to a multi-armed bandit system using the Upper Confidence Bound algorithm (UCB1) that selects among competing content topics or training scenario job families, combined with a deadline pressure mechanism called Caffeine Mode that dynamically modulates exploration-exploitation balance based on real-time schedule deficit.

BACKGROUND OF THE INVENTION

In autonomous content generation systems for social media, selecting which topic or content type to generate next is a sequential decision problem under uncertainty. A content system that always selects the historically best-performing topic (pure exploitation) misses the opportunity to discover new high-performing topics. Conversely, a system that randomly explores topics uniformly (pure exploration) wastes publication opportunities on known underperformers. The optimal solution is a principled explore-exploit algorithm that balances both objectives.

The UCB1 algorithm (Auer et al., 2002) provides a mathematically optimal solution to the multi-armed bandit problem: each arm is assigned a score equal to its empirical average reward plus an exploration bonus proportional to the square root of the logarithm of total trials divided by arm-specific trials. This score is proven to achieve logarithmic regret – the lowest possible regret growth rate for the stochastic multi-armed bandit problem.

A further challenge arises in deadline-driven content systems: when an autonomous content scheduler falls behind its posting schedule, it must close the deficit as quickly as possible. Under deadline pressure, the optimal strategy shifts from

balanced explore-exploit toward aggressive exploitation of known high-yield domains, and the timing window for posting must widen to accept suboptimal posting times. Prior art systems do not provide a mathematically grounded mechanism for continuously adapting exploration-exploitation balance as a function of real-time schedule deficit.

A similar challenge exists in machine learning training scenario selection: when the training loop falls behind its year-equivalent step target, the system must shift toward known high-yield training domains (those associated with the highest gradient quality) and reduce time spent on exploratory scenario families that have not yet demonstrated value.

There exists a need for a unified bandit selection framework that (1) applies UCB1 with a calibrated exploration constant suited to the reward signal range of the target domain, (2) continuously computes a schedule pressure metric from real-time posting counts or training step deficits, (3) modulates the UCB1 exploration bonus as a function of this pressure, and (4) identifies high-yield arm categories that receive an explicit exploitation bonus under critical pressure.

SUMMARY OF THE INVENTION

The present invention provides a UCB1 multi-armed bandit selection system comprising per-arm reward tracking using an incremental mean estimator, an exploration bonus scaled by a calibrated constant, a real-time schedule pressure computation engine, a Caffeine Mode activation mechanism that applies an exploitation bias to identified high-yield arms when pressure exceeds a threshold, and an interval scheduler that reduces the next-generation interval under increasing pressure.

In a first aspect, the invention provides a method of selecting a topic or domain from a plurality of candidates comprising: computing a UCB1 score for each candidate as the sum of the

candidate's average reward and an exploration bonus; selecting the candidate with the highest UCB1 score; and updating the selected candidate's average reward and trial count using an incremental mean estimator after observing the outcome.

In a second aspect, the invention provides a Caffeine Mode pressure engine that computes a dimensionless pressure score from the ratio of posts or training steps remaining to time remaining, classifies the pressure into four tiers (on-track, mild lag, behind, critical), and for each tier applies a distinct modification to exploration weight, posting timing windows, and next-generation interval.

In a third aspect, the invention provides a high-yield arm set that receives an additive UCB1 score bonus proportional to the current pressure level, causing the bandit to shift toward known high-yield arms under critical deadline pressure without entirely abandoning the explore-exploit guarantee.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

I. UCB1 Multi-Armed Bandit for Topic Selection

The content generation system maintains two maps for each known topic t :

topicPerformanceMap(t): the empirical average engagement rate of published content on topic t

topicTrialCountMap(t): the number of times topic t has been selected and its outcome observed

The UCB1 score for topic t is computed as:

$$UCB1(t) = \text{avg_reward}(t) + UCB1_C \times \sqrt{(\ln(N) / n(t))}$$

where N = total trials across all topics, $n(t)$ = trials for topic t , and $UCB1_C = 0.25$. The exploration constant $UCB1_C = 0.25$ is calibrated for engagement rate reward signals in the range $[0, 1]$, specifically for social media engagement rates

that typically fall between 0.001 and 0.10. At this constant, the exploration bonus at $n(t) = 1$ and $N = 10$ is $0.25 \times \sqrt{(2.303)} \approx 0.379$, sufficient to ensure every topic is tried at least once in the first 10 rounds regardless of the observed reward of early-tried topics.

The topic with the highest UCB1 score is selected for each content generation cycle. After publication, the observed engagement rate is recorded and the incremental mean is updated:

$$\begin{aligned} \text{avg_reward}(t) &\leftarrow (\text{avg_reward}(t) \times n(t) + \text{new_reward}) / (n(t) + 1) \\ n(t) &\leftarrow n(t) + 1 \end{aligned}$$

II. UCB1 Multi-Armed Bandit for Training Scenario Families

The same UCB1 algorithm is applied to scenario job family selection in the AB Test Scenario Layer. The arms are the eight job families of the Music Industry Scenario Engine. The reward signal is a composite of compound depth (normalized) and scenario intensity:

$$\text{reward} = \min(1.0, \text{depth_reward} + \text{intensity} \times 0.3)$$

where $\text{depth_reward} = 0.2$ (depth 0), 0.5 (depth 1), or 0.8 (depth 2+).

The exploration bonus uses the same $\text{UCB1_C} = 0.25$ constant and an approximation of $\sqrt{N/n}$ in place of $\sqrt{\ln(N)/n}$ for numerical stability with small trial counts.

III. Caffeine Mode Pressure Engine

The schedule pressure for content generation is computed from posting history:

$$\text{pressure} = \frac{\max(0, (\text{min_posts_per_day} - \text{posts_today}))}{\max(0.5, \text{hours_remaining})}$$

This produces a dimensionless ratio with units of posts per hour. Four pressure tiers are defined:

Tier 0 (on-track): pressure = 0 → normal operation

Tier 1 (mild lag): $0 < \text{pressure} \leq 0.5$ → minor posting window relaxation

Tier 2 (behind): $0.5 < \text{pressure} \leq 1.5$ → gate floor -7 points, minInterval posting

Tier 3 (critical): pressure > 1.5 → CAFFEINE MODE: gate floor -10 points, 20-min interval

The pressure tier is broadcast to downstream systems only when it changes, preventing redundant updates during sustained pressure states.

For training scenario selection, the pressure is derived from the year-equivalent deficit of the time simulator:

$$\text{pressure} = \min(1.5, \text{deficit_years} / \text{MAX_YE_DEFICIT} \times 1.5)$$

where MAX_YE_DEFICIT is a configurable saturation point (default 5.0 simulated years).

IV. Caffeine Mode Modifications

Under each pressure tier, the following modifications are applied:

Content generation:

- The content quality gate threshold is reduced by pressure-proportional amount (see Patent No. 4)
- The next content generation interval is reduced from 2 hours (base) to 30 minutes (mild), 20 minutes (critical)
- Optimal timing window constraints are bypassed: content is published whenever it is ready rather than waiting for the statistically optimal time-of-day

Scenario selection:

- High-YE job families (release_architect, visual_director, fan_engagement) receive an additive UCB1 score bonus = $\min(0.5, \text{pressure} / 3.0)$

- This bonus shifts the bandit toward exploitation of proven high-gradient families without entirely eliminating exploration of untried families

V. Platform Rotation

For content generation across multiple social media platforms, a deterministic round-robin platform selector maintains a rotation index per user and returns platforms in sequence. This ensures that no platform is systematically neglected during normal operation. Under Caffeine Mode, the rotation may be accelerated so that the platform with the greatest posting deficit ($\text{min_posts_per_day} - \text{posts_today}$) is prioritized.

VI. Next-Generation Interval Scheduler

The interval scheduler computes the next content generation interval as follows:

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if pressure > 1.5: return caffeineModeInterval (20 minutes)
if pressure > 0.5: return minInterval (30 minutes)
if avg_engagement > 0.05: return minInterval
if avg_engagement < 0.01: return maxInterval (6 hours)
else: return baseInterval (2 hours)
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This produces a feedback loop where high-performing content in normal mode accelerates posting, while schedule pressure independently triggers acceleration regardless of engagement performance.

VII. Pressure Broadcasting

The pressure computation is executed before each content generation cycle. The result is compared to the last broadcast pressure tier; if the tier has changed, the new pressure value is broadcast to:

- The content quality pipeline (updates floor score and urgency scoring bonuses)

- The HyperLearning engine (applies deadline pressure to learning multiplier)
- The AB Test Scenario Layer (adjusts UCB1 exploration-exploitation balance)

Broadcasting occurs only on tier changes, not on continuous pressure value changes, to avoid flooding downstream systems with redundant updates.

VIII. Fully Deterministic Selection

A critical implementation detail is that UCB1 selection is fully deterministic: given the same topicPerformanceMap, topicTrialCountMap, and total N, the selected topic is always the same. No random number generation is used in the selection step. This property is important for auditability: operators can reproduce the exact sequence of topic selections from the logged performance data.

For cold-start (empty performance map), the default topic is selected using a seeded hash of the user ID, producing a consistent default topic for each user across restarts.

CLAIMS

1. A computer-implemented system for selecting among a plurality of candidate topics or domains for content generation or model training, the system comprising: a processor; and a non-transitory computer-readable medium storing instructions that, when executed by the processor, cause the processor to: maintain for each candidate an empirical average reward value and a trial count; compute a UCB1 score for each candidate as the sum of the candidate's average reward and a product of a configurable exploration constant and the square root of the logarithm of total trials divided by candidate-specific trials; select the candidate with the highest UCB1 score; update the selected candidate's average reward using an incremental mean estimator after observing an outcome; and compute a real-time schedule pressure as a ratio of remaining required outputs to remaining available time, and modify the UCB1 score of identified high-yield candidates by adding a pressure-proportional exploitation bonus.
2. The system of claim 1, wherein the exploration constant is 0.25, calibrated for reward signals in the range from zero to one representing engagement rate metrics for social media content.
3. The system of claim 1, wherein the pressure-proportional exploitation bonus applied to high-yield candidates is bounded by a maximum value and increases linearly from zero at zero pressure to the maximum value at a critical pressure threshold.
4. The system of claim 1, wherein the schedule pressure is classified into four tiers comprising on-track, mild lag, behind, and critical, and wherein each tier triggers a distinct set of downstream modifications to content generation interval, quality gate threshold, and posting

timing window constraints.

5. The system of claim 4, wherein the critical pressure tier triggers a Caffeine Mode state in which: the content generation interval is reduced to a minimum crunch interval; optimal posting time-of-day constraints are bypassed; and the quality gate floor threshold is reduced by a maximum allowable decrement.
6. The system of claim 1, further comprising a broadcasting mechanism that propagates the schedule pressure value to a plurality of downstream systems only when the pressure tier changes, preventing redundant updates during sustained pressure states.
7. The system of claim 1, wherein the content schedule pressure is computed as the ratio of the number of posts required to meet a daily minimum minus posts already published today, divided by the maximum of a minimum time floor and the number of hours remaining in the current day.
8. The system of claim 1, wherein the training scenario pressure is derived from a year-equivalent step deficit computed by a training time simulator as the gap between a target year-equivalent throughput and actual accumulated year-equivalent steps.
9. The system of claim 1, wherein the candidate selection step is fully deterministic such that, given identical per-candidate reward maps and trial counts, the same candidate is always selected, enabling auditability by reproduction of the selection sequence from logged reward data.
10. The system of claim 1, wherein a cold-start default is selected using a seeded hash of a user or session

identifier, producing a consistent initial candidate selection for each user independently of the random state of the execution environment.

11. The system of claim 1, further comprising a next-generation interval scheduler that returns a minimum interval under high schedule pressure, a maximum interval when average engagement falls below a lower threshold, and an intermediate base interval under normal conditions.
12. The system of claim 11, wherein the next-generation interval at the critical pressure tier is shorter than the next-generation interval applied when recent average engagement exceeds a high-performance threshold, such that schedule pressure can override engagement-based acceleration.
13. A method of adaptively selecting a domain for artificial intelligence model training under deadline pressure, the method comprising: maintaining empirical average reward values and trial counts for each of a plurality of training domains; selecting a training domain by maximizing a UCB1 score comprising an exploitation term and an exploration term; computing a training deficit pressure as a normalized measure of how far actual training throughput lags behind a target throughput schedule; applying an additive exploitation bonus to UCB1 scores of domains identified as high-yield when the training deficit pressure exceeds a threshold; and recording the reward earned by training on the selected domain and updating the domain's empirical average reward using an incremental mean estimator.
14. The method of claim 13, wherein the reward signal for a training domain is a composite function of the compound depth of scenarios drawn from that domain and the

intensity of the highest-scoring scenario variant produced by a quality gate evaluation.

15. The method of claim 13, further comprising reducing a quality threshold required to accept a scenario for training when the training deficit pressure exceeds a moderate threshold, while maintaining an absolute minimum quality floor independent of pressure level.
16. The method of claim 13, wherein the exploitation bonus increases linearly with training deficit pressure from zero at zero pressure to a maximum bonus value at a critical pressure saturation point.
17. A non-transitory computer-readable medium storing instructions that, when executed by a processor, implement: a UCB1 bandit engine configured to maintain empirical average rewards and trial counts for each of a plurality of selectable arms, compute UCB1 scores, and select the arm with the highest score; a pressure computation engine configured to derive a dimensionless pressure score from a real-time deficit between required and actual outputs; a high-yield arm registry storing identifiers of arms that receive an additive UCB1 score bonus proportional to the current pressure score; a tier classifier configured to classify the pressure score into a plurality of discrete tiers and trigger distinct behavioral modifications for each tier; and a pressure broadcaster configured to propagate the pressure score to registered downstream systems only on tier transitions.
18. The computer-readable medium of claim 17, further storing instructions that implement a platform rotation selector that returns social media platforms in a deterministic round-robin sequence, ensuring even coverage of all configured platforms during normal operation.

19. The computer-readable medium of claim 17, wherein the UCB1 bandit engine updates arm rewards using a Welford incremental mean estimator that maintains a running average without storing the full history of observed rewards.
20. The computer-readable medium of claim 17, further storing instructions that implement a performance adaptation engine that, in addition to UCB1 selection, adjusts posting frequency limits and optimal timing windows based on aggregate platform-specific engagement trends derived from a rolling history of recent content performance observations.

ABSTRACT

A UCB1 multi-armed bandit selection system selects among candidate content topics or training scenario domains by computing a score equal to empirical average reward plus an exploration bonus with exploration constant 0.25, selecting the highest-scoring candidate deterministically, and updating reward estimates using a Welford incremental mean estimator. A Caffeine Mode pressure engine computes a real-time schedule pressure from the ratio of outstanding required outputs to remaining available time, classifies this into four tiers, and for each tier applies distinct modifications to content generation intervals, quality gate thresholds, and posting timing constraints. Under critical pressure, identified high-yield candidates receive an additive UCB1 score bonus proportional to the pressure level, shifting the bandit toward exploitation of proven performers without eliminating exploration. The pressure value is broadcast to a plurality of downstream systems only on tier transitions. The selection mechanism is fully deterministic, enabling auditability by reproduction of the selection sequence from logged reward data.